

KOYUK, AK, USA

WMO: 702223

Lat: 64.935N

Lon: 161.155W

Elev: 34

StdP: 100.92

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 26602

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.3	(c) -28.8	(d) -34.0	(e) 0.2	(f) -26.2	(g) -33.2	(h) 0.2	(i) -26.3	(j) 11.2	(k) -16.7	(l) 10.3	(m) -17.2	(n) 1.2	(o) 50	(p) 0.774

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.3	(c) 23.0	(d) 14.2	(e) 21.2	(f) 14.0	(g) 18.7	(h) 13.0	(i) 15.9	(j) 20.9	(k) 14.7	(l) 19.3	(m) 13.6	(n) 17.8	(o) 3.2	(p) 10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	9.7	17.6	12.5	9.0	16.5	11.4	8.4	15.4	44.5	21.0	41.0	19.2	38.0	17.7	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.3	7.6	-33.2	27.0	4.7	2.3	-36.7	28.7	-39.4	30.0	-42.1	31.3	-45.5	33.0		
			-31.3	16.9	2.9	1.2	-33.4	17.8	-35.0	18.5	-36.7	19.2	-38.7	20.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.4	-13.7	-12.6	-11.9	-4.2	4.4	11.6	13.4	11.7	6.9	-0.9	-9.2	-13.0
	DBStd	11.93	9.00	8.47	7.05	6.51	5.47	3.90	3.36	3.25	3.72	4.67	6.36	8.02
	HDD10.0	4446	736	634	679	425	186	26	7	18	104	339	577	714
	HDD18.3	7214	994	867	938	675	431	204	158	207	343	598	827	973
	CDD10.0	281	0	0	0	0	13	73	112	71	12	1	0	0
	CDD18.3	8	0	0	0	0	1	2	4	2	0	0	0	0
	CDH23.3	66	0	0	0	0	6	19	33	7	0	0	0	0
	CDH26.7	11	0	0	0	0	0	3	7	1	0	0	0	0
Wind	WSAvg	3.6	4.3	4.3	4.1	3.8	3.2	3.1	3.1	3.1	3.2	3.1	3.5	3.9
Precipitation	PrecAvg	522	34	33	27	22	26	34	63	84	67	53	42	37
	PrecMax	687	76	112	78	51	50	56	122	167	124	136	90	78
	PrecMin	333	8	8	5	7	8	15	10	23	29	16	9	15
	PrecStd	88	17	21	17	12	11	12	27	27	26	28	22	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	3.0	2.9	10.2	23.8	27.0	27.1	23.9	18.0	11.4	5.9	4.8
		MCWB	2.0	0.2	-0.2	4.5	12.8	15.1	16.4	16.4	10.8	7.4	3.4	1.8
	2%	DB	2.4	2.1	1.3	7.8	18.0	22.7	23.0	21.2	15.9	7.8	2.8	2.4
		MCWB	0.3	-0.3	-1.1	2.8	10.4	13.3	14.7	14.3	10.0	5.3	0.8	0.1
	5%	DB	0.9	0.2	-0.9	6.2	14.5	20.2	21.2	18.6	13.0	6.3	1.3	-0.1
		MCWB	-1.0	-1.4	-2.4	1.7	8.0	13.0	14.8	13.2	9.4	4.1	-0.3	-1.5
	10%	DB	-2.0	-2.0	-2.3	3.9	12.3	17.7	18.6	17.1	12.2	5.0	0.1	-2.3
		MCWB	-3.3	-3.5	-4.0	0.3	6.9	11.5	13.8	12.5	9.1	3.3	-1.0	-3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	0.8	0.5	4.7	13.0	16.2	18.0	17.2	11.9	7.7	3.9	2.4
		MCDB	5.4	2.8	2.4	9.2	22.8	21.3	24.2	23.3	15.1	10.4	5.4	3.5
	2%	WB	0.5	-0.3	-0.8	3.2	10.8	14.7	16.4	15.4	10.8	5.8	1.4	0.2
		MCDB	2.1	1.5	0.9	7.2	17.4	20.9	21.6	19.3	13.6	7.5	2.5	2.1
	5%	WB	-1.1	-1.4	-2.2	1.9	8.9	13.3	15.1	14.1	10.2	4.6	-0.1	-1.5
		MCDB	0.2	0.1	-0.7	5.6	13.7	18.7	19.7	17.4	12.9	6.4	1.0	-0.3
	10%	WB	-3.0	-3.2	-3.8	0.8	7.3	12.3	14.0	13.1	9.2	3.4	-1.4	-3.5
		MCDB	-1.7	-1.8	-2.3	3.7	11.6	17.1	18.1	16.1	11.8	4.9	-0.1	-2.2
Mean Daily Temperature Range	5% DB	MDBR	5.2	6.0	6.6	7.3	8.0	7.8	6.3	6.4	6.2	4.8	4.6	5.0
		MCDBR	6.7	5.6	6.7	7.8	12.2	11.6	10.8	9.6	8.0	5.2	4.3	5.6
		MCWBR	5.7	4.6	5.5	5.3	7.3	6.3	5.9	5.4	5.0	4.2	4.0	4.8
	5% WB	MCDBR	6.5	5.7	6.6	7.1	11.3	9.7	9.3	7.7	6.9	5.0	4.4	5.3
		MCWBR	5.7	5.2	5.5	4.9	6.9	5.7	5.5	4.8	4.5	4.2	4.3	4.8
Clear-Sky Solar Irradiance	taub	0.178	0.255	0.269	0.342	0.359	0.356	0.363	0.384	0.313	0.273	N/A	0.181	
	taud	1.909	2.146	2.239	2.086	2.274	2.400	2.410	2.349	2.530	2.380	N/A	1.911	
	Ebn at Noon	351	642	820	814	843	857	838	775	779	635	N/A	97	
	Edh at Noon	23	57	81	123	113	102	99	96	64	45	N/A	8	
All-Sky Solar Radiation	RadAvg	0.15	0.77	2.53	4.49	5.24	5.61	4.53	3.45	2.17	0.90	0.23	0.04	
	RadStd	0.02	0.09	0.21	0.27	0.50	0.51	0.32	0.42	0.23	0.11	0.03	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page